Whistlestop Coffee Hut

UML Database Design

> **Project:** Whistlestop Coffee Hut - CSC8019 Team 6
> **Database:** Aiven MySQL
> **ORM:** Prisma
> **Version:** 1.0
> **Date:** April 2026

Project Overview

Whistlestop Coffee Hut is a comprehensive coffee ordering system that allows users to:

- Browse menu items
- View individual menu item details
- Add items to shopping cart
- Register and login
- View customer pages (loyalty rewards, order status, etc.)

- Access staff pages (dashboard, archive, etc.)

Core Entity Design

1. User

Purpose: Stores customer and staff user information

Key Fields:

- `id` - Unique identifier (CUID)
- `email` - Unique email address
- `role` - User role (CUSTOMER, STAFF, ADMIN)
- `loyaltyPoints` - Accumulated loyalty points
- `loyaltyStamps` - Collected stamps

Relationships:

- Has multiple CartItems
- Places multiple Orders
- Has LoyaltyRecords
- Receives Notifications

2. Station

Purpose: Represents each coffee station location

Key Fields:

- `id` - Unique identifier
- `slug` - URL-friendly identifier
- `location` - Physical location details
- `isActive` - Operational status

Relationships:

- Has multiple OpeningHours
- Offers StationMenuItems
- Fulfills multiple Orders

3. MenuItem

Purpose: Core product offerings

Key Fields:

- `name` - Item name
- `slug` - URL identifier
- `category` - Product category
- `priceRegular / priceLarge` - Pricing by size
- `isAvailable` - Availability status

Categories: COFFEE, TEA, PASTRY, SNACK, COLD_DRINK, CHOCOLATE

4. Order

Purpose: Represents customer orders

Key Fields:

- `userId` - Customer who placed order
- `stationId` - Pickup location
- `status` - Order status
- `totalAmount` - Final order cost
- `pickupTime` - Scheduled pickup time
- `isArchived` - Staff archive status

Status Flow: PENDING → ACCEPTED → PREPARING → READY → COLLECTED

5. PaymentRecord

Purpose: Payment transaction history

Key Fields:

- `orderId` - Associated order (unique)
- `provider` - Payment provider
- `amount` - Payment amount
- `status` - PENDING, PAID, FAILED, REFUNDED
- `transactionRef` - Provider's transaction ID

6. LoyaltyRecord

Purpose: Tracks loyalty points and stamp transactions

Transaction Types:

- `EARN` - Earn points
- `REDEEM` - Redeem points
- `ADJUSTMENT` - Manual adjustment

Key Design Features

1. Historical Data Preservation

`OrderItem` stores `nameAtTime` and `priceAtTime` to maintain historical accuracy. Even if menu items are updated, order history remains accurate.

2. Soft Deletions

Orders can be `isArchived` instead of deleted. This provides audit trail and data recovery.

3. Location-Specific Inventory

`StationMenuItem` allows per-station stock and availability control. Different stations can have different inventories.

4. Flexible Relationships

`Notification` 's `userId` and `orderId` are optional, supporting system-wide or order-specific notifications.

5. Audit Trail

> `AuditLog` tracks all important operations, supporting compliance and debugging.

Database Constraints

Entity	Constraint	Purpose
`User`	`email` unique	Prevent duplicate accounts
`MenuItem`	`slug` unique	Enable URL routing
`Station`	`slug` unique	Enable URL routing
`CartItem`	`[userId, menuItemId, size]` unique	One cart item per user/item/size
`StationMenuItem`	`[stationId, menuItemId]` unique	One record per station/item
`PaymentRecord`	`orderId` unique	One payment per order
`OpeningHour`	`[stationId, dayOfWeek]` unique	One record per station/day

Technology Stack

Component	Technology
Database	MySQL (Aiven Cloud)
ORM	Prisma
Framework	Next.js
Language	JavaScript/TypeScript

Component	Technology
Authentication	JWT Session-based
UI Components	shadcn/ui
Styling	Tailwind CSS
Icons	Lucide React

All Entities List

1. User
2. Station
3. MenuItem
4. StationMenuItem
5. CartItem
6. Order
7. OrderItem
8. LoyaltyRecord
9. PaymentRecord
10. OpeningHour
11. TrainSchedule
12. TrainSelection
13. Notification
14. AuditLog

🎯 **ER Diagram and UML Class Diagram**

For complete ER diagram and UML class diagram, please refer to the `UML_DATABASE_DESIGN.md` file in the project

How to Use This Documentation

1. **Understand Database Structure:** Read the description of each entity to understand its purpose and key fields
2. **View Relationships:** Understand the relationships and constraints between entities

3. **Reference Design Principles:** Learn about design features such as historical data preservation and soft deletions
4. **Consult Complete Diagrams:** Refer to the ER diagram and UML class diagram in the Markdown version

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